From Zero-Shot to Fine-Tuning: Measuring the Domain Gap and the Role of Attention in Motorcyclist Helmet Detection

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***Abstract*—Automatic motorcyclist helmet detection from traffic cameras supports road-safety enforcement. Two common assumptions in practice are that COCO pre-trained weights are directly usable and that adding attention mechanisms improves accuracy. This study tests both assumptions under controlled conditions using a public dataset of 1,803 images containing three classes (helmet, license plate, and motorcyclist). We first measure the domain gap using a zero-shot evaluation on COCO weights. We then conduct a controlled multi-seed evaluation ($n = 3$) of three identically fine-tuned architectures: YOLOv8s (pure CNN), YOLO11s (with C2PSA self-attention), and YOLOv8s augmented with CBAM. The zero-shot evaluation confirms a massive domain gap: mAP@0.5 is only 0.0986–0.1143 and the helmet and plate classes are undetected. Fine-tuning closes this gap, boosting mAP@0.5 to approximately 0.96 and showing that domain adaptation is far more critical than architectural complexity. However, attention mechanisms yield no statistically significant advantage over the pure CNN. YOLOv8s reaches 0.9596 ± 0.0008 mAP@0.5 at 296 FPS. YOLO11s does not differ significantly from YOLOv8s ($p = 0.11$, $p = 0.37$), and adding CBAM reduces mAP@0.5 significantly ($p = 0.004$). These results demonstrate that added attention does not guarantee performance gains on small-scale helmet-detection datasets.**

***Index Terms*—helmet detection, object detection, attention mechanism, YOLOv8, YOLO11, CBAM, transfer learning, domain gap**

I. Introduction

Motorcycles account for a large share of traffic-accident casualties, especially in countries where they dominate daily mobility. The helmet is the most decisive protective factor against fatal head injuries, and monitoring compliance at scale requires automation. Manual observation does not scale to real traffic volumes, whereas automatic object detection from camera footage can operate continuously and consistently [1].

The YOLO family has become the de facto standard for real-time detection, with each iteration improving the backbone, neck, and label-assignment strategy [2]–[5]. YOLO11 is the most recent iteration and introduces positional self-attention blocks (C2PSA) into the feature pathway, making it a natural successor to the attention-free YOLOv8. Separately, lightweight attention modules such as CBAM [6] or Squeeze-and-Excitation (SE) networks [7] are a popular way to inject channel-spatial attention into an existing CNN backbone without replacing the architecture entirely.

Several recent studies inject attention modules into YOLO-family detectors and report accuracy improvements on helmet-detection datasets [8]–[10]. Those studies, however, run on different datasets, base architectures, and training protocols, so their numbers cannot be compared directly with ours. They also generally compare a single configuration without repetition across seeds and omit statistical significance tests, so the reported gains could stem from hyperparameters, augmentation, or initialisation rather than from the attention mechanism itself.

A second assumption is rarely verified: that COCO pre-trained weights are already close to the helmet task. The helmet class schema (helmet, license plate, motorcyclist) only partly overlaps with the 80 COCO classes [21], and two of the three classes have no counterpart at all. The size of this domain gap should be measured, not assumed.

We address both gaps with a controlled evaluation. The contribution is methodological rather than a claim that attention is harmful: an identical protocol, multi-seed repetition, formal statistical testing, and an explicit separation between the domain gap (the zero-shot stage) and the attention effect (the fine-tuning stage). Two research questions follow:

- **RQ1.** How large is the domain gap between COCO weights as-is and the helmet-detection task?
- **RQ2.** After fine-tuning with an identical protocol and multi-seed repetition, do attention mechanisms (C2PSA in YOLO11s, CBAM in YOLOv8s) surpass the pure CNN YOLOv8s in a statistically meaningful way?

II. Literature Review

A. Helmet Detection

Research on helmet detection has evolved from simple classification toward pipelines that detect, track, and classify motorcycles together with their riders. Siebert and Lin [1] demonstrated large-scale helmet-use detection from traffic video using a single-stage detector and released a widely used annotated dataset. In the enforcement context, several pipelines combine helmet detection with license-plate localisation and recognition to identify violators [8], [11], [12].

A growing body of work integrates attention mechanisms into helmet-detection models, commonly by adding channel-spatial attention modules to YOLO-family detectors or inserting feature-fusion blocks together with attention layers [9], [10]. These studies typically report accuracy improvements over their respective baselines, but they generally compare a single configuration without repetition across seeds, use protocols tuned per model, and omit statistical significance tests. This methodological gap motivates the controlled, multi-seed comparison presented here.

B. YOLO Architectures and Attention Modules

Single-stage detectors formulate detection as direct regression of bounding boxes and classes in a single network pass, trading some accuracy for high speed and employing specialised loss functions such as Focal Loss [13] to mitigate class imbalance [2]. The YOLO family has become the de facto standard for real-time detection [3]. YOLOv8 represents a mature CNN generation without explicit attention modules, relying on efficient convolutional blocks, Feature Pyramid Networks (FPN) [14], and Path Aggregation Networks (PANet) [15] to handle multi-scale objects. YOLO11 introduces C2PSA, a positional self-attention block embedded into the feature pathway, which places it at an intermediate point between a pure CNN and a full transformer design.

Surveys of attention in vision [16], [17] conclude that the effectiveness of attention mechanisms depends heavily on training-data volume and task complexity. On the lightweight side, modules such as CBAM [6] and Coordinate Attention [18] sequentially apply channel and spatial attention to emphasise both what is relevant and where it appears in a feature map. They add a small number of parameters and can be inserted into any CNN without replacing the backbone. For small-object detection, Cheng et al. [19] observe that objects such as helmets and license plates within wide frames require multi-scale feature representations and high input resolution.

III. Methodology

A. Dataset

We use a public helmet-detection dataset from Roboflow Universe (NCKH 2023 Helmet Detection Project, version 19, MIT license). The dataset contains 1,803 annotated images in YOLO format with three classes: helmet, license plate, and motorcyclist. A fixed split of 1,563 training (86.7%), 140 validation (7.8%), and 100 test images (5.5%) was used throughout. All final metrics were computed on the test split, which was never seen during training. The dataset is daytime-oriented with adequate lighting and limited occlusion, conditions that are relatively favourable compared with real CCTV footage.

The dataset comprises 8,229 annotation instances with an imbalanced class distribution, summarised in Table I. The license-plate class is the most frequent (42.5%), while the helmet class is both the least frequent (24.0%) and the smallest in size, reinforcing its status as the hardest class. Beyond export preprocessing (auto-orientation with EXIF stripping and resizing to 1280×720 by stretching), the dataset applies a three-versions-per-source augmentation: random rotation of $\pm 5°$, shear of $\pm 5°$, brightness adjustment of $\pm 20\%$, and salt-and-pepper noise on 5% of pixels. Additional training-time augmentation (mosaic, HSV, horizontal flip, scale) is described in Section III-C.

TABLE I
Image and annotation-instance distribution per split.

Split	Images	Instances	helmet	plate	m.cyclist
Train	1,563	7,100	1,682	3,039	2,379
Validation	140	671	177	268	226
Test	100	458	117	190	151
Total	**1,803**	**8,229**	**1,976**	**3,497**	**2,756**

TABLE II
The three architectures and their attention mechanism.

Model	Attention	Params
YOLOv8s	None (baseline)	$\sim$11.17 M
YOLO11s	Partial (C2PSA)	$\sim$9.4 M
YOLOv8s+CBAM	Channel + spatial	$\sim$11.51 M

Across the 100 test images, the distribution of annotation instances reflects a dominance of the `motorcyclist` and `helmet` classes as a natural pair, whereas the `license_plate` class appears less frequently due to specific camera viewpoints. This natural class imbalance is a characteristic of the domain that should be considered when interpreting aggregate mAP metrics.

Three architectures were chosen to occupy different positions on the attention spectrum, as summarised in Table II. YOLOv8s is the pure-CNN baseline with no explicit attention. YOLO11s adds partial self-attention through C2PSA blocks while remaining a single-stage detector. YOLOv8s+CBAM applies a controlled attention intervention by inserting CBAM modules into the baseline network. To handle the natural class imbalance inherent in the helmet-detection dataset, all three models rely on YOLO's native task-aligned loss formulation. This consists of Complete Intersection over Union (CIoU) loss and Distribution Focal Loss (DFL) for bounding box regression, combined with Binary Cross-Entropy (BCE) loss for classification. The loss is joint-optimized using a Task-Aligned Assigner (`TaskAlignedAssigner`) with default hyperparameter weights (cls=0.5, box=7.5, and dfl=1.5) to align classification scores with localization accuracy, mitigating class imbalance effects without manual resampling.

Three CBAM modules were inserted at the outputs of the three detection scales (P3, P4, P5) just before the detection head, so the YOLOv8s+CBAM variant is identical to the plain YOLOv8s except for the added attention. The addition raises the parameter count from 11.17 to 11.51 million, an increase of about three percent. This small increment isolates the effect of attention while keeping all other factors constant. The CBAM variant was tested in a single configuration only, namely this P3/P4/P5 placement with a single learning rate and no separate warmup for the attention module; these are acknowledged implementation limitations.

B. Two Evaluation Protocols

Protocol 1, zero-shot COCO baseline. The COCO pre-trained weights of YOLOv8s and YOLO11s are evaluated on the test split without any training, as a verification of the domain-gap size. Because COCO recognises 80 classes that include neither helmet nor license plate, predictions are mapped to our three-class schema: COCO motorcycle (class ID 3) becomes motorcyclist, while other categories such as person (class ID 0) are deliberately ignored to focus specifically on the target motorcyclist class and avoid false positive pedestrian detections. Meanwhile, helmet and license plate have no counterpart and therefore score zero by design. Metrics are computed with `pycocotools` at a low confidence threshold (0.001) so the precision-recall curve is correct. This protocol does not compare architectures meaningfully, since both fail on two of three classes; it confirms that fine-tuning is mandatory.

Protocol 2, fine-tuning. All three architectures are trained from COCO pre-trained weights [21] with an identical protocol using the Ultralytics framework (version 8.4.65) [20] on PyTorch (version 2.8.0+cu128) with a single NVIDIA RTX 4090 GPU (24GB VRAM) running inside Windows Subsystem for Linux (WSL2 on Ubuntu, Python 3.12.13, CUDA 12.8). For the CBAM variant, the attention layers were randomly initialised while the rest of the network inherited COCO weights. The unified configuration used an input resolution of 1280 pixels, automatic batch-size selection (which resolved to an optimal training batch size of 6 via Ultralytics AutoBatch at approximately 54% CUDA memory utilization), automatic optimizer selection (AdamW, lr=0.001429), an initial learning rate of 0.01, identical augmentation (mosaic, HSV shifting, horizontal flipping, scaling), and early stopping with patience 25 over a maximum of 100 epochs. Each architecture was trained on three seeds (42, 0, 1) run sequentially and independently, with GPU memory cleared between runs. The 1280-pixel resolution, rather than the 640-pixel default, was motivated by the small relative size of the helmet and license-plate classes, consistent with Cheng et al. [19].

C. Evaluation Metrics and Statistical Testing

The primary metrics are mean Average Precision (mAP) at an IoU threshold of 0.5 (mAP@0.5) and averaged over thresholds from 0.5 to 0.95 (mAP@[.5:.95]), supplemented by inference speed in frames per second (FPS). Average Precision for a class is the area under the precision-recall curve:

$$AP = \int_0^1 P(R)\, dR \tag{1}$$

TABLE III
ZERO-SHOT COCO BASELINE ON THE TEST SPLIT ($n = 1$).

Model	**mAP@0.5**	**mAP@.5:.95**	**helmet**	**plate**	**m.cyclist**
YOLOv8s	0.0986	0.0270	0.000	0.000	0.296
YOLO11s	0.1143	0.0320	0.000	0.000	0.343

where $P(R)$ denotes precision as a function of recall R. The mAP is the arithmetic mean of AP across the set of target classes C:

$$mAP = \frac{1}{|C|} \sum_{c \in C} AP_c \tag{2}$$

FPS is reported as an indicative trade-off measure, not a controlled benchmark, since it was measured on a single dedicated run with batch size 1 while the GPU was idle.

To assess accuracy differences for the fine-tuned models, we paired results between models on the same seed and applied a paired t-test to the per-seed metric difference, with significance level $\alpha = 0.05$. The paired t-statistic is:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \tag{3}$$

where $\bar{d}$ is the sample mean of the paired differences, s_d their sample standard deviation, and n the number of pairs ($n = 3$ seeds). We also report Cohen's d for paired samples:

$$d = \frac{\bar{d}}{s_d} \tag{4}$$

Pairing by seed controls for initialisation variation and increases sensitivity to architectural effects [22]. With $n = 3$, the test has limited power; p-values and effect sizes are presented as supplementary information, with the awareness that only large differences can be detected convincingly.

IV. Results

A. Domain-Gap Verification (RQ1)

Table III summarises the zero-shot evaluation. COCO models as-is cannot be used for helmet detection: mAP@0.5 is only 0.0986 (YOLOv8s) and 0.1143 (YOLO11s). This aggregate is dominated by the motorcyclist class, the only class with a COCO counterpart (AP@0.5 of 0.296 and 0.343), while helmet and license plate are zero by design. Fig. 1 shows the per-class breakdown, and Fig. 2 contrasts the zero-shot and fine-tuned regimes. The small difference between the two architectures here is not meaningful, since both fail on two of three classes; the zero-shot stage serves only to confirm that fine-tuning is mandatory. This answers RQ1: the domain gap is large.

B. Fine-Tuned Performance (RQ2)

After fine-tuning, accuracy jumps to about 0.96 mAP@0.5 and closes the domain gap. Table IV summarises the multi-seed results. On mAP@0.5, YOLOv8s achieves the highest value together with the best inference speed. YOLO11s is comparable but does not exceed the baseline, and

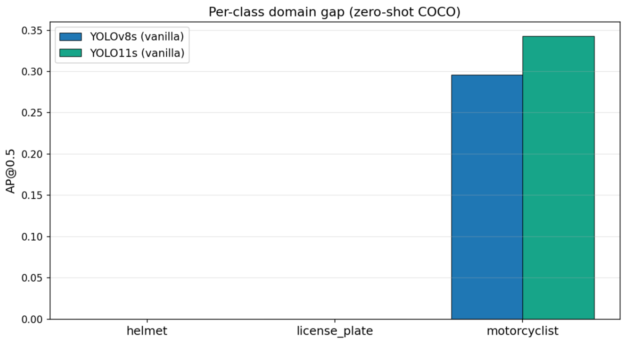

Fig. 1. Per-class AP@0.5 under zero-shot COCO. The helmet and license-plate classes are zero by design (no COCO counterpart); only the motorcyclist class receives a signal through the mapping from COCO motorcycle.

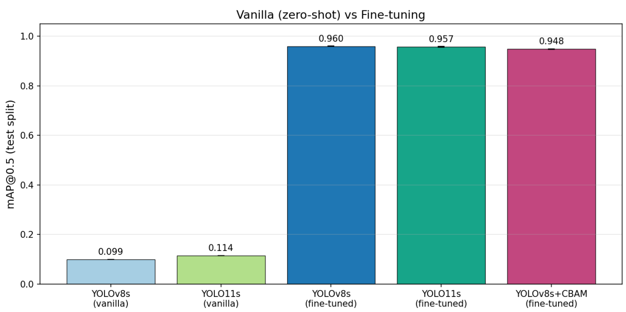

Fig. 2. Accuracy jump from the zero-shot baseline (about 0.10) to the fine-tuned models (about 0.96) on mAP@0.5. The difference among architectures after fine-tuning is far smaller than the effect of fine-tuning itself.

TABLE IV
FINE-TUNED PERFORMANCE ON THE TEST SPLIT (MEAN $\pm$ STD, $n = 3$ SEEDS).

Model	mAP@0.5	mAP@.5:.95	FPS
YOLOv8s	$\mathbf{0.9596 \pm 0.0008}$	0.6885 ± 0.0006	$\sim$ **296**
YOLO11s	0.9574 ± 0.0019	$\mathbf{0.6902 \pm 0.0022}$	~ 189
YOLOv8s+CBAM	0.9481 ± 0.0006	0.6832 ± 0.0034	~ 290

YOLOv8s+CBAM records the lowest mAP@0.5. On the stricter mAP@[.5:.95], the order reverses: YOLO11s slightly leads YOLOv8s, while CBAM remains lowest. Fig. 3 visualises the mAP@0.5 comparison with error bars showing the standard deviation across seeds.

C. Significance Testing

Table V reports the paired t-tests on both metrics. On mAP@0.5, adding CBAM reduces accuracy significantly and consistently ($\Delta = +0.0116$ for the baseline, $t(2) = 16.1$, $p = 0.004$, $d \approx 9.3$); CBAM is also lower than YOLO11s ($p = 0.012$). The YOLOv8s versus YOLO11s difference is small (0.0022) and not significant ($t(2) = 2.71$, $p = 0.11$). On mAP@[.5:.95], no pair is significant, including YOLOv8s versus CBAM ($p = 0.079$). This reversal across metrics confirms that the YOLOv8s advantage is not consistent across all evaluation dimensions. This answers RQ2: no attention

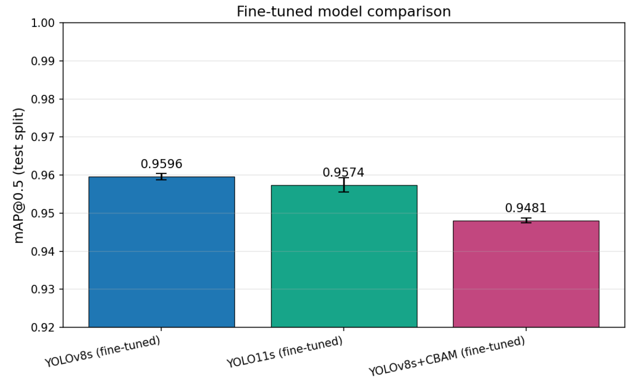

Fig. 3. mAP@0.5 across the three fine-tuned architectures (mean $\pm$ std, $n = 3$). YOLOv8s leads slightly on mAP@0.5; on mAP@[.5:.95] the order changes (YOLO11s highest, Table IV).

TABLE V
PAIRED T-TEST RESULTS ACROSS SEEDS.

Comparison	Δ	t(2)	p	Cohen's d
mAP@0.5				
YOLOv8s vs CBAM	+0.0116	16.1	0.004	9.3
YOLOv8s vs YOLO11s	+0.0022	2.71	0.11	1.56
YOLO11s vs CBAM	+0.0093	8.92	0.012	5.15
mAP@[.5:.95]				
YOLO11s vs YOLOv8s	+0.0018	1.16	0.37	0.67
YOLOv8s vs CBAM	+0.0053	3.33	0.079	1.92
YOLO11s vs CBAM	+0.0071	2.52	0.13	1.45

mechanism meaningfully surpasses the pure CNN, and CBAM in the configuration tested is significantly detrimental on mAP@0.5.

D. Qualitative Analysis

Fig. 4 shows a typical detection output from the fine-tuned YOLOv8s on a test image. The rider, helmet, and license plate are detected simultaneously with tight bounding boxes. Visual inspection of the test sample confirms that remaining errors concentrate on very small or partially occluded helmets, consistent with the per-class behaviour observed under zero-shot conditions.

V. DISCUSSION

The jump from about 0.10 (zero-shot) to about 0.96 (fine-tuned) confirms that COCO weights as-is are far from ready for helmet detection, and two of the three classes have no counterpart concept in COCO. The practical implication is that every deployment must budget for fine-tuning on domain data, and any performance claim must be reported after domain adaptation rather than from a zero-shot test.

Among the fine-tuned models, no attention is consistently superior across all metrics. YOLOv8s leads on mAP@0.5, but YOLO11s is slightly ahead on mAP@[.5:.95], and the two do not differ significantly. This pattern aligns with surveys showing that the benefit of attention depends on dataset scale and heterogeneity [16], [17]. On a small and relatively easy

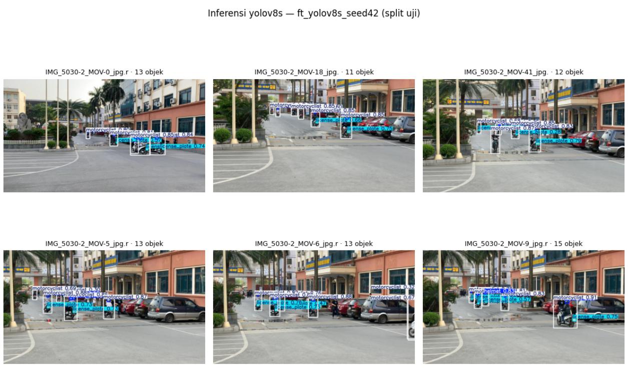

Fig. 4. Example detection output from the fine-tuned YOLOv8s on a test image. All three classes are detected simultaneously.

dataset where the baseline already reaches about 0.96, the room for improvement is thin and insufficient to reveal an attention advantage.

Adding CBAM in the configuration tested reduces mAP@0.5 significantly. Several explanations cannot be ruled out. The randomly initialised CBAM blocks may disturb the mature feature flow inherited from COCO when trained with a single learning rate and no separate warmup; the placement at P3/P4/P5 may be suboptimal; and the ceiling effect near 0.96 makes even a small decrease appear consistent and significant. The very small standard deviation of the CBAM variant (± 0.0006) confirms the reduction is stable across seeds. This result does not generalise to all CBAM implementations: a different placement, a separate learning rate, training from scratch, or a warmup schedule may produce different outcomes. It is a result for one specific configuration, not a general statement about CBAM.

The findings apply specifically to a small, relatively clean dataset and a training protocol optimised for YOLO. Under more challenging conditions, such as nighttime CCTV footage with heavy occlusion and high traffic density, the capacity of attention to model global context may prove beneficial. Within this scope, YOLOv8s remains the most rational choice for real-time helmet detection on similar data. Because a system that links helmet detection to license plates can identify individuals, deployment within enforcement infrastructure also requires an appropriate regulatory and ethical framework.

VI. Threats to Validity

The study uses a single small dataset that is curated, well-lit, and lightly occluded, so generalisation to harder data, where attention may genuinely help, has not been verified. With $n = 3$ seeds and a baseline near the 0.96 ceiling, the paired t-test has low power; the equivalence of YOLO11s and the baseline should be read as an absence of evidence of a difference, not as evidence of equivalence, and $p = 0.11$ is not sufficient for an equivalence claim. Hyperparameters were not tuned per architecture, so a protocol tuned for YOLO may disadvantage YOLO11s or the CBAM variant. Finally, the CBAM modules were inserted only at the detection head; placement in the backbone, training from scratch, or a separate learning rate for the attention parameters might yield different results, so the observed reduction cannot be attributed solely to the attention module itself.

VII. Conclusion

We separated two sources of improvement that are often mixed in the helmet-detection literature. The domain gap from the COCO starting point is large (mAP@0.5 about 0.10 zero-shot), so fine-tuning is mandatory. After fine-tuning under an identical protocol with multi-seed evaluation and paired t-tests, no attention mechanism consistently surpasses the pure CNN: YOLOv8s leads on mAP@0.5 (0.9596) at the fastest speed (~ 296 FPS), YOLO11s is slightly ahead on mAP@[.5:.95] (0.6902) without significance, and CBAM in the configuration tested significantly reduces mAP@0.5 ($p = 0.004$). These findings caution against assuming that newer or more complex architectures automatically outperform simpler baselines on small-scale helmet-detection tasks.

Future work should test generalisation on larger and more challenging datasets, including real CCTV footage, increase the number of seeds with an a priori power analysis, and run a systematic CBAM ablation varying placement, training duration, and learning-rate allocation to clarify whether the observed reduction is inherent or an artefact of the experimental design.

Statements

Data Availability. The dataset is publicly available through Roboflow Universe as the NCKH 2023 Helmet Detection Project, version 19, under the MIT license. The training scripts, configurations, and experiment notebooks are available in the authors' research repository.

Conflict of Interest. The authors declare no conflict of interest.

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